

Data: /net/vega/data/users/observatory/images/YYMMDD/

- .../ST7/i or /g (old ccd)
- .../STL-6303E/i or /g (main imager)

(from the knowledge database)

- Data Reduction Basic Steps
  - Visit the page [Data Reduction Cheat Sheet](#)
- Further for this project
  - You need to align all of your images – but do not combine them
  - For each filter the you are assigned, you will do relative photometry (measuring brightness) over time to create a light curve
  - Identify other, non-variable, stars in the field to use as reference
- Further Analysis for this project
  - Plot the corrected magnitude of the star as a function of time for each filter – note that you should get the observation time from the header of the image, and account for the length of the exposure!
  - (You can also investigate if the colour between bands changes over time)

Can you answer these questions with your work?

- What is the shape of the variability? Is it sinusoidal or some other shape?
- What is the period of the variability (the time between successive peaks or dips)?
- What is the total amplitude of the variability? Does it change with band?
- Does the star's colour change with time?
- What kind of star is this? Why do these sort of stars pulsate?

<https://blauwobserverknowledgebase.readthedocs.io/en/latest/pm/a1-scientific-poster.html#a1-poster>

<https://blauwobserverknowledgebase.readthedocs.io/en/latest/pm/a2-scientific-talk.html#a2-talk>